

Final Assignment

Section 1: Programming Tasks

- **Task 1 (Basic Class):** Create a class named **Laptop** with properties for **brand**, **processor**, and **price**. Create an object of this class and print all its details.
- **Task 2 (Method Implementation):** Create a class **Circle** that takes **radius** as a parameter in the `__init__` method. Add a method called **calculated_area** that returns the area of the circle (Formula: πr^2) and a method called **calculated_perimeter** that returns the perimeter of the circle (Formula: $2\pi r$).
- **Task 3 (Object Manipulation):** Create a **Player** class with **name** and **score**. Create an object, print the initial score, update the score with new value, and print the updated score.
- **Task 4 (Conditional Logic):** Create a **Temperature** class. If the input temperature is above 37°C, the method **check_health** should print "Fever", otherwise "Normal".
- **Task 5 (Inheritance):** Create a parent class **Vehicle** with a method **move()**. Create a child class **Airplane** that inherits from **Vehicle** and adds a unique method **fly()**.

Section 2: Scenario-based Challenges

- **ATM System:** Create a class named **Class**. The account should contain: Account number, Account holder name, Balance. Implement the following methods: **deposit()**, **withdraw()**, **show_balance()**. The system must:
 - Allow users to deposit and withdraw money.
 - Prevent withdrawal when the balance is insufficient.
 - Display the current balance.
 - Create at least two different accounts.
 - Perform transactions on both accounts.

- **Library Management System:** Create two classes named **Book** and **Library**. In **Book** class, each book should contain: Title, Author, ISBN, Availability Status. In **Library** class, the library should provide methods to: **add_method()**, **borrow_book()**, **return_book()**, **display_books()**. The system should:
 - Allow books to be added to the library.
 - Display all available books.
 - Allow user to borrow a book
 - Prevent a book from being borrowed if it is already borrowed.
 - Allow borrowed books to be returned.
 - Update the availability status automatically.
- **University Hierarchy:** Create a **Person** parent class. Derive two classes from it: **Teacher** (with a **subject** property) and **Student** (with a **cgpa** property). Use **super().__init__()** to initialize the common properties in both child classes.

Optional Task: (It may add extra marks)

- **E-Commerce Order System:**

Create the following classes:

- Product
- Customer
- Order

Product should contain: Product name, Price, Quantity, and method **total_price()**.

Customer should contain: Customer name, Customer ID

Order should:

- Contain customer name
- Contain multiple products.
- Calculate the subtotal.
- Apply a discount.
- Calculate the final price.
- Methods are: **add_products()**, **subtotal()**, **discount()**, **final_price()**, **display_order()**

Discount Rule: If the subtotal is greater than 10,000, apply a 10% discount. Otherwise, no discount should be applied.

Requirements: Create a customer, add multiple products to an order, and display: Customer Name, Products, Subtotal, Discount, Final Price.